

SGT UNIVERSITY

School of Engineering & Technology (SOET)

Developer Specification Document Attendance, Mentorship, Leave Arrangement, Extra Class & Detention Monitoring App

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Purpose: App design, coding, database planning, workflow implementation and dashboard/report development

Core Design Principle

The mentor is the key protagonist of the system. The app must not merely collect attendance; it must enable mentors to identify risk early, counsel students, regularize approved cases, communicate with parents, issue warning letters through proper approval, and be monitored by HoD/Dean for accountability.

1. Project Objective

Develop a role-based, web/mobile-friendly app for SOET, SGT University to manage classroom attendance, activity attendance, mentor-led attendance regularization, faculty leave/class arrangement, extra/make-up classes, detention list generation, parent communication, warning letters, and Dean/HoD-level monitoring.

2. Guiding Business Requirements

The system must ensure that teachers only see their assigned subjects, mentors only see their assigned mentees, and HoDs/Dean can monitor attendance, compliance, mentor performance, pending approvals, detention risk, and defaulting mentors. All attendance changes must maintain audit trails and approval logs.

3. Recommended Technology Approach

The system may be developed as a responsive web app with optional mobile PWA support. Suggested stack: React/Next.js or Angular for frontend, Node.js/NestJS or Django/FastAPI for backend, PostgreSQL/MySQL for database, role-based authentication, REST/GraphQL APIs, and PDF/Excel generation for reports and warning letters.

4. User Roles and Access Control

Role	Access / Responsibility
Admin	Configure academic year, programmes, semesters, sections, subjects, timetable, faculty, students, mentor mapping, HoD/Dean mapping, attendance rules, detention threshold, templates and approval workflows.
Teacher	View only assigned subjects/classes; mark regular attendance; mark no class conducted; schedule extra/make-up classes for assigned batches; apply leave and arrange replacement teachers.
Mentor	View only assigned mentees; see subject-wise attendance; initiate approved regularization; add counselling notes; generate parent summaries; initiate warning letters; maintain mentor intervention history.
Activity Coordinator	Create approved activity attendance for seminars, workshops, industrial visits, placement events, expert lectures, university duties etc.; submit participation for HoD approval.
Arrangement Teacher	Accept/reject arrangement class requests; mark attendance for accepted arrangement classes.
HoD	Approve leave/arrangements, mentor regularization, activity attendance, extra-class policies where enabled; monitor department attendance, mentor performance, detained lists and warning actions.
Dean	School-level monitoring of all departments; view attendance trends, detention risk, pending approvals, defaulting mentors, HoD compliance, reports and escalation dashboards.

5. Core App Modules

Module	Purpose
Dashboard	Role-specific overview with KPIs, alerts, pending actions and quick links.
Subject Attendance	Teacher marks attendance only for assigned subject/section/slot.
Mentee Monitor	Mentor sees assigned mentees only, with subject-wise color-coded attendance and intervention actions.
Leave & Arrangement	Teacher applies leave and selects available teachers for affected lectures; arrangement teacher accepts; HoD approves.
Extra / Make-Up Classes	Teacher schedules extra class from available batch slots; attendance counts for student detention and teacher course completion.
Activity Attendance	Coordinator marks attendance for approved academic activities; HoD approves.
Regularization	Mentor submits sick leave, internship, project, university duty or approved academic duty regularization for mentees.
Detention & Warning	Generate detention risk list, defaulter list, warning letters, parent summaries, and escalation records.
Reports	Excel/PDF reports for teacher, mentor, HoD, Dean and parent-facing summaries.
Settings / Admin	Master data, timetable, thresholds, templates, permissions, academic calendar and audit rules.

6. Master Data and Database Entities

Entity	Important Fields
Student Master	Student ID, roll no., name, programme, semester, section, batch, email, mobile, parent/guardian contact, mentor ID, active status.
Faculty Master	Faculty ID, name, department, designation, email, mobile, role flags, teaching competency tags, active status.
Subject Master	Subject code, name, credits, programme, semester, section, planned lecture count, course plan topics, assigned teacher.
Timetable	Date/weekday, slot ID, start/end time, programme, semester, section, subject, teacher, room/lab, lecture type.
Mentor Mapping	Mentor ID, student IDs, effective dates, mapped by Admin/HoD, status.
Attendance Transaction	Student ID, subject, date, slot, status, marked by, class type, source, approval status, remarks, timestamp.
Leave & Arrangement	Leave request, affected slots, selected arrangement teachers, acceptance status, HoD approval.
Extra Class	Subject, batch/section, date, free slot, reason, topic covered, attendance status, course completion mapping.
Activity Attendance	Activity type, coordinator, date/slot, participants, proof/document, approval status.
Mentorship Record	Counselling notes, parent calls, warning issued, student response, improvement plan, next review date.
Warning Letter	Student, mentor, reason, attendance data, letter stage, approval, issue date, parent copy status.
Audit Log	Action, old value, new value, user, role, timestamp, IP/device, approval trail.

7. Attendance Business Rules

Rule	Requirement
Assigned subject visibility	Teacher shall see and select only subjects officially assigned to him/her. No global subject list should be visible to teachers.
Assigned mentee visibility	Mentor shall see only assigned mentees, even if mentees belong to different batches/programmes.
Regular attendance	Marked by subject teacher for regular class only. Counts for student attendance and teacher course completion.
No Class Conducted	Marked by teacher when regular class is not conducted. Does not count for student attendance, detention calculation, or teacher course completion.
Arrangement class	Taken by arrangement teacher during original teacher leave/unavailability. Counts for student attendance and detention calculation; does not count for original teacher course completion.
Extra / make-up class	Taken by original subject teacher in available batch slot. Counts for student attendance, detention calculation and teacher course completion.
Activity attendance	Marked by activity coordinator for approved academic activity; counts only after HoD approval and must be mapped to date/slot.
Mentor regularization	Mentor may initiate for sick leave, internship, project, university duty, approved academic duty etc.; counts only after HoD approval and as per university rule.
No duplicate attendance	A student cannot get two attendance credits for the same date/slot/subject. System must prevent duplicate records.
Audit trail	Every submit, edit, approval, rejection, reopening and correction must be logged.

8. Teacher Login Functional Requirements

Teacher login should be fast, simple and limited to assigned responsibilities. The teacher should not have to search through all subjects or all students.

Feature	Developer Instruction
Assigned Subject Selection	Display subject cards/dropdown containing only assigned subject-section combinations. Auto-fill programme, semester and section.
Attendance Marking	Allow Present, Absent, Remarks, Save Draft, Submit Attendance, Mark No Class Conducted. Lock after defined time unless reopened by HoD/Admin.
Arrangement Class Visibility	If a teacher accepted an arrangement class, show it in his/her dashboard as Arrangement Class and allow attendance marking.
Leave Application	Show leave form; auto-list affected lectures; system should show available replacement teachers for each affected slot based on free timetable.
Extra Class Scheduling	Allow teacher to select assigned subject, available batch free slot, reason, topic covered and class type. Extra class attendance counts for students and course completion.
Course Completion	Show regular classes conducted, no class conducted, arrangement classes, extra classes, planned

9. Mentor-Centric Requirements

Mentor as Key Protagonist

The mentor dashboard is the central intervention layer. Detention prevention, parent communication, warning letters and student counselling should begin from the mentor dashboard. HoD and Dean dashboards must measure mentor responsiveness and default.

Mentor Feature	Requirement
Assigned Mentee Table	Show only assigned mentees with subject headers and corresponding attendance percentages.
Color Coding	Green: safe; Amber: warning; Red: critical; Grey: pending approval/no data. Thresholds configurable by Admin.
Subject Drill-down	Click any subject attendance to view total classes held, attended, absent, arrangement attendance, activity attendance, approved regularization and pending cases.
Mentorship Notes	Mentor can record counselling note, student reason, corrective action, next review date, and student response.
Parent Communication	Mentor can record phone/email/meeting communication with parent/guardian, attach summary, and schedule follow-up.
Regularization Request	Mentor can submit approved category attendance regularization with proof/remarks for HoD approval.
Parent Summary Report	Generate parent-friendly attendance summary for each mentee with subject-wise status, shortage, action taken and improvement advice.
Warning Letter Initiation	Mentor can initiate warning letter for students below threshold after defined number of interventions.
Mentor Default Tracking	System must track if mentor has not acted on low-attendance mentees within the defined timeline. This must be visible to HoD and Dean.

10. Leave and Class Arrangement Workflow

Step	System Behaviour
1. Teacher applies leave	Teacher selects leave type, date range and reason. System identifies affected lectures.
2. Available teachers shown	For each affected lecture, system lists only teachers free in that slot and preferably from matching department/competency.
3. Arrangement teacher accepts/rejects	Selected teacher receives request and accepts or rejects. If rejected, original teacher selects another teacher.
4. HoD approval	HoD sees leave application with class arrangement status. HoD approves/rejects leave.
5. Attendance marking	Accepted arrangement teacher marks attendance. Attendance counts for student attendance/detention but not original teacher course completion.
6. Reporting	Arrangement classes appear in separate reports for workload support and monitoring.

11. Extra / Make-Up Class Workflow

Step	System Behaviour
1. Teacher identifies backlog	Dashboard shows no class conducted, holiday/event loss and syllabus/course completion gap.
2. Slot selection	Teacher selects from pre-fed available slots for the specific batch/section. Prevent clashes with other classes, exams or approved activities.
3. Topic mapping	Teacher enters topic covered and reason such as syllabus backlog, no class compensation or slow progress.
4. Attendance marking	Teacher marks attendance after class.
5. Count logic	Attendance counts for student attendance/detention and teacher course completion. No Class Conducted is excluded until compensated.

12. Activity Attendance Workflow

Activity Type	Handling
Seminar / Workshop / Expert Lecture	Coordinator creates event, selects participants, maps slots, uploads proof and submits for HoD approval.
Industrial Visit	Coordinator marks students participating; teacher marks No Class Conducted due to approved activity where relevant.
Placement / Training / Hackathon	Attendance credited only to participants and only after approval.
Project / Internship	Mentor initiates or validates student-specific status; HoD approves.
University Duty	Mentor/coordinator submits proof; HoD approves.

13. Detention List Generation

Detention Calculation Principle

The detention list must be generated using valid attendance after accounting for regular classes, arrangement classes, extra classes, approved activity attendance, and approved mentor regularization. No Class Conducted must be excluded from the denominator and from course completion unless compensated by extra class.

Attendance Source	Counted for Student Attendance / Detention?	Counted for Teacher Course Completion?
Regular class marked by assigned teacher	Yes	Yes
Arrangement class marked by accepted arrangement teacher	Yes	No for original teacher; visible as arrangement duty for substitute teacher
Extra / make-up class by original subject teacher	Yes	Yes
Approved activity attendance	Yes	No, unless mapped by policy as academic contact hour
Approved internship/project/university duty	Yes, as per policy	No
Sick leave	Configurable as per university rule; must be separately tagged	No
No Class Conducted	No; exclude from denominator	No

14. Detention List Views for Teacher, HoD and Dean

User	Detention View / Report
Teacher	Subject-wise shortage list for subjects taught by the teacher; students below threshold; attended vs held classes; impact of extra classes; no class conducted count; export option.
Mentor	Mentee-wise detention risk list across all subjects; warning level; parent communication status; counselling status; pending regularization; next intervention due.
HoD	Department/programme/semester/section-wise detained and at-risk list; subject-wise shortage; teacher compliance; mentor action status; pending approvals; warning letters issued.
Dean	School-wide detention dashboard with department comparison, programme trend, mentor default list, HoD pending approvals, warning letters, parent report status and critical students.

Detention Report Columns	Details
Student Details	Name, roll no., programme, semester, section, mentor name, parent contact.
Attendance Details	Subject-wise attended, valid classes held, attendance %, shortage classes required to reach threshold.
Attendance Breakup	Regular, arrangement, extra class, approved activity, approved regularization, absent, pending approval.
Risk Level	Safe / Watch / Warning / Critical / Detention Recommended. Thresholds configurable.
Mentor Action	Counselling done, parent informed, warning letter issued, regularization pending, next follow-up date.
Approvals	Pending HoD approvals, rejected cases, document status.
Export	Excel and PDF, with filters by department, programme, semester, subject, mentor, risk status.

15. Parent Attendance Summary Report Through Mentors

The system shall generate parent-facing attendance summary reports routed through mentors. The report must be simple, non-technical and suitable for email/WhatsApp/PDF sharing as per university policy.

Report Section	Fields
Student Profile	Name, roll no., programme, semester, section, mentor name and contact.
Overall Attendance	Overall percentage, status, minimum required percentage, shortage classes.
Subject-wise Attendance	Subject, classes held, attended, approved credits, absences, percentage, risk color.
Mentor Intervention	Counselling dates, advice given, student response, improvement plan.
Parent Action Required	Suggested actions: ensure regular attendance, meet mentor/HoD, submit valid documents if applicable.
Next Review Date	Date for follow-up by mentor.
Disclaimer	Attendance is subject to final verification and university rules.

16. Warning Letters and Escalation Workflow

Warning Stage	Trigger and Workflow
Stage 1: Advisory Alert	Student falls below configurable threshold, e.g., 75% or risk band. System alerts mentor. Mentor counsels student and records note.
Stage 2: Parent Intimation	Student remains below threshold after defined days/classes. Mentor generates parent summary and records parent communication.
Stage 3: Formal Warning Letter	Mentor initiates warning letter. HoD reviews and approves. Letter issued to student and parent/guardian.
Stage 4: Critical Detention Risk	Student remains critical despite warning. HoD/Dean dashboard flags case for review.
Stage 5: Detention Recommendation	Final list generated after approval workflow and pending regularizations are resolved or marked rejected.

Warning Letter Template Fields	Description
Letter No. / Date	Auto-generated.
Student Information	Name, roll no., programme, semester, section.
Attendance Summary	Overall and subject-wise percentage with shortage.
Prior Interventions	Counselling dates, parent contact, previous warning if any.
Required Compliance	Minimum attendance required and corrective action.
Signatory	Mentor prepared by, HoD approved by, Dean visibility where required.
Delivery Tracking	Email/SMS/printed copy, parent copy status, acknowledgement.

17. Mentor Accountability and Defaulting Mentor Dashboard

Dean-Level Requirement

Dean must be able to see which mentor is defaulting. A defaulting mentor is one who has assigned mentees in risk/critical attendance bands but has not completed required interventions within the defined time window.

Default Metric	Definition
Unreviewed At-Risk Mentees	Number of mentees below threshold for whom mentor has not opened/reviewed case.
Counselling Pending	Number of at-risk mentees without counselling note within X days of alert.
Parent Contact Pending	Number of warning/critical mentees where parent communication is not recorded within X days.
Regularization Pending at Mentor Level	Cases not submitted/updated by mentor despite student proof or pending category.
Warning Letter Pending	Eligible students where mentor has not initiated warning letter.
Follow-up Overdue	Cases where next review date has passed and no update is recorded.
Mentor Compliance Score	Composite score based on timely review, counselling, parent contact, regularization closure and warning actions.

Dean / HoD Mentor Dashboard Columns	Details
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Mentor Name	Faculty mentor name and department.
Assigned Mentees	Total assigned mentees.
At-Risk Mentees	Number below threshold / warning / critical.
Action Taken	Counselling done, parent contact done, warning letters initiated.
Pending Actions	Counselling pending, parent contact pending, warning pending, follow-up overdue.
Default Status	On Track / Watch / Defaulting / Critical Default.
Compliance Score	Percentage or points-based score.
Action	View details, send reminder, escalate to HoD, download report.

18. Role-Based Dashboard Requirements

Dashboard	Must Show
Teacher Dashboard	Today classes, pending attendance, assigned subjects, assigned mentees, quick actions, course completion, extra class backlog, leave arrangement status.
Mentor Dashboard	Assigned mentees, risk classification, subject-wise shortages, pending interventions, parent reports, warning letters, counselling notes.
HoD Dashboard	Department attendance, detained/at-risk students, pending approvals, teacher compliance, mentor compliance, leave/arrangement, extra classes, warning letters.
Dean Dashboard	School-level summary, department comparison, detention risk, pending HoD approvals, defaulting mentors, high-risk students, parent communication status, warning letters and trend analytics.

19. Required Reports and Exports

Report	Users	Purpose
Daily Attendance Report	Teacher/HoD/Dean	Date-wise subject and student attendance.
Subject-wise Attendance Report	Teacher/HoD	Classes held, attended, shortage, no class conducted, extra classes.
Student-wise Attendance Report	Mentor/HoD/Dean	Complete attendance breakup by subject and source.
Mentee Summary Report	Mentor	Assigned mentee list with risk status and actions.
Parent Attendance Summary	Mentor	Parent-facing PDF/email summary.
Warning Letter Register	Mentor/HoD/Dean	All warning letters initiated, approved, issued and acknowledged.
Detention Risk List	Teacher/Mentor/HoD/Dean	Students below threshold or likely to be detained.
Final Detention List	HoD/Dean	Final list after regularization and approvals.
Mentor Default Report	HoD/Dean	Mentors defaulting on intervention timelines.
Teacher Compliance Report	HoD/Dean	Attendance marking status, no class conducted, extra classes, leave arrangements.
Activity Attendance Report	HoD/Dean	Activity-wise attendance and approvals.
Course Completion Report	Teacher/HoD/Dean	Planned lectures, regular classes, extra classes, no class conducted, percentage completion.

20. Alerts and Notification Rules

Trigger	Notification To
Teacher has not marked attendance within deadline	Teacher, HoD; repeated default visible to Dean.
Student below warning threshold	Mentor and student; parent only as per escalation rule.
Student critical below threshold	Mentor, HoD and Dean dashboard alert.
Mentor action overdue	Mentor reminder; visible to HoD and Dean as default.
Parent communication pending	Mentor reminder; HoD visibility after defined time.
Warning letter eligible but not initiated	Mentor and HoD alert; Dean dashboard count.
Regularization pending approval	HoD reminder; Dean view for long-pending cases.
Arrangement teacher acceptance pending	Arrangement teacher and original teacher; HoD visibility.
Extra class scheduled but attendance not marked	Teacher reminder; HoD visibility.

21. UI/UX Design Instructions

UI Area	Instruction
Design Style	Gen Z-friendly minimal design: clean white background, pastel accents, rounded cards, light shadows, simple icons and uncluttered tables.
Role Clarity	Each screen should clearly show user role and scope: Teacher, Mentor, HoD or Dean.
Restricted Data Visibility	Clearly indicate "Only assigned subjects shown" and "Only assigned mentees shown."
Color Coding	Use consistent green/amber/red/grey status colors across dashboard, reports and warning workflows.
Mobile Support	Attendance marking, mentee review and approvals should be mobile-responsive.
Fast Actions	Mentor dashboard must include quick actions: counselling note, parent contact, regularization, warning letter, parent report.
Accessibility	Use readable fonts, high contrast, keyboard-friendly inputs and clear status text along with colors.

22. Approval and Permission Matrix

Action	Teacher	Mentor	Coordinator	HoD	Dean	Admin
Mark regular attendance	Yes, assigned only	No	No	View/Reopen	View	Configure
Mark arrangement attendance	If accepted	No	No	View	View	Configure
Schedule extra class	Yes, assigned only	No	No	Approve/View if enabled	View	Configure
Submit mentor regularization	No	Yes, mentees only	No	Approve/Reject	View	Configure

Submit activity attendance	No	No	Yes	Approve/Reject	View	Configure
Generate parent summary	No	Yes	No	View	View summary	Template configure
Initiate warning letter	No	Yes	No	Approve/Issue	View/Escalate	Template configure
Generate final detention list	View assigned subject	View mentees	No	Generate Dept	Generate School	Configure rules
View defaulting mentors	No	No	No	Dept-level	School-level	No

23. Attendance Calculation Logic

Use the following simplified logic; exact rules should be parameterized in Admin Settings.

Metric	Formula / Logic
Valid Classes Held for Subject	Regular classes conducted + arrangement classes + extra classes + approved activity slots where applicable - no class conducted exclusions.
Student Attended Count	Present in regular + present in arrangement + present in extra + approved activity + approved regularization credits.
Attendance %	$\text{Student Attended Count} / \text{Valid Classes Held} \times 100$.
Shortage Classes	Minimum additional attended classes needed to reach configured threshold.
Teacher Course Completion Count	Regular classes taken by original teacher + extra/make-up classes taken by original teacher. Exclude no class conducted and arrangement classes.
Mentor Compliance Score	Weighted score based on intervention timeliness, counselling, parent contact, regularization closure, warning actions and follow-up closure.

24. Non-Functional Requirements

Area	Requirement
Security	Role-based access control, encrypted passwords, HTTPS, secure session management, audit trail.
Performance	Attendance list should load quickly for 100+ students; dashboard summaries should be cached or precomputed where required.
Scalability	Architecture should support all SOET departments and future expansion to other schools/faculties.
Auditability	All changes and approvals must be traceable for disputes and compliance.
Data Export	Excel and PDF exports required for all major reports.
Data Import	Admin should be able to bulk upload students, faculty, timetable, subjects and mentor mapping using Excel templates.
Backup	Daily database backup and versioned report storage.
Privacy	Parent/student contact data visible only to authorized users.

25. Acceptance Criteria for Developer

- Teacher can view and mark attendance only for assigned subjects.
- Mentor can view only assigned mentees and subject-wise color-coded attendance.
- Leave module auto-lists affected lectures and shows only free arrangement teachers for each slot.
- Arrangement teacher acceptance is required before/alongside HoD leave approval.
- Arrangement attendance counts for student detention but not original teacher course completion.
- Extra class slot selection uses available batch slots and counts for both student attendance and teacher course completion.
- Activity attendance is credited only after approval and cannot duplicate slot attendance.
- Teacher, mentor, HoD and Dean can generate detention-related views appropriate to role.
- Mentor can generate parent summaries and initiate warning letters.
- Dean can see defaulting mentors and drill down to mentor-wise pending interventions.
- All key reports can be exported to Excel/PDF.
- All edits, approvals and rejections have audit trail.

26. Suggested Development Milestones

Phase	Deliverables
Phase 1: Master Data & Authentication	Role login, permissions, master data and mappings.
Phase 2: Attendance Core	Assigned subject attendance, locking and reports.
Phase 3: Mentor Module	Mentee dashboard, counselling, parent log and regularization.
Phase 4: Leave, Arrangement & Extra Classes	Leave workflow, arrangement attendance and extra class scheduling.
Phase 5: Detention, Warning & Parent Reports	Detention lists, warnings, parent summaries and mentor accountability.
Phase 6: HoD/Dean Dashboards	Monitoring, mentor defaulting dashboard, analytics and exports.
Phase 7: QA, Security & Deployment	Testing, security review, deployment and training.